UNIT 4 PRACTICAL REPORT

Deploying and Connecting to a Windows Server Instance on Amazon EC2 — with Evidence

Student	Vishal Shinde
Cloud platform / Region	AWS — Asia Pacific (Mumbai), ap-south-1
Operating system	Windows Server 2025
Instance type	t3.micro

1. Objective

To launch a Windows Server virtual machine on Amazon EC2, configure restricted Remote Desktop Protocol (RDP) access, verify network connectivity, troubleshoot a failed connection, and connect to the Windows instance.

2. Instance configuration

Parameter	Value
Instance name	Unit4-Windows-Server-New
Instance ID	i-0ee4b1191425bf835
Public IPv4 (during practical)	16.4.34.160
Private IPv4	172.31.2.247
AMI / OS	Windows Server 2025 — English Full Base
Instance type / root disk	t3.micro / 30 GiB gp3
VPC / Subnet	vpc-0dced0a92e1375770 / subnet-0fd367b02c8162311
Security group	launch-wizard-3 — sg-0e346fbac42bb8209

3. Procedure performed

1. Launched a new EC2 instance using the Windows Server 2025 base image in the Mumbai region.

2. Verified the instance was Running and the system, instance, and EBS status checks passed.

3. Configured the security group to allow RDP (TCP 3389) only from a specific public IP address using a /32 CIDR.

4. Checked the subnet Network ACL and confirmed inbound and outbound allow rules were present.

5. Checked the route table and confirmed the default route (0.0.0.0/0) to an Internet Gateway was active.

6. Ran Test-NetConnection to TCP port 3389. The first test failed on the original network.

7. Switched the laptop to a mobile hotspot and updated the RDP security-group source to the hotspot public IP 223.238.161.207/32.

8. Repeated Test-NetConnection; TcpTestSucceeded returned True on the hotspot.

9. Opened Remote Desktop Connection to 16.4.34.160 and proceeded with the Administrator credentials. The user confirmed the connection succeeded.

4. Test result and conclusion

Test	Result
Initial TCP 3389 test	TcpTestSucceeded: False on original network
TCP 3389 test on mobile hotspot	TcpTestSucceeded: True after updating the RDP source IP
Final outcome	User confirmed successful Windows Remote Desktop connection

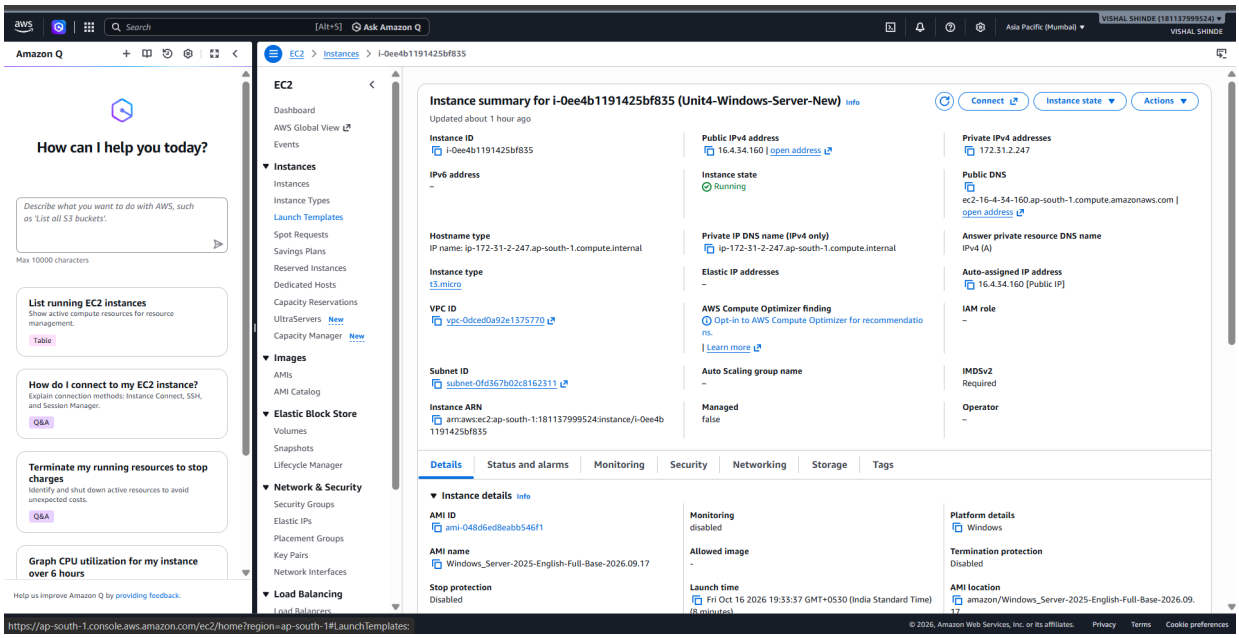
5. Security notes

RDP was restricted to a single trusted public IP at a time; it was not opened to 0.0.0.0/0. If the hotspot public IP changes, update the security-group source. Never share the Administrator password or private key (.pem). Stop the instance when it is not needed to reduce compute charges; attached EBS storage may still incur charges while stopped.

APPENDIX — SCREENSHOT EVIDENCE

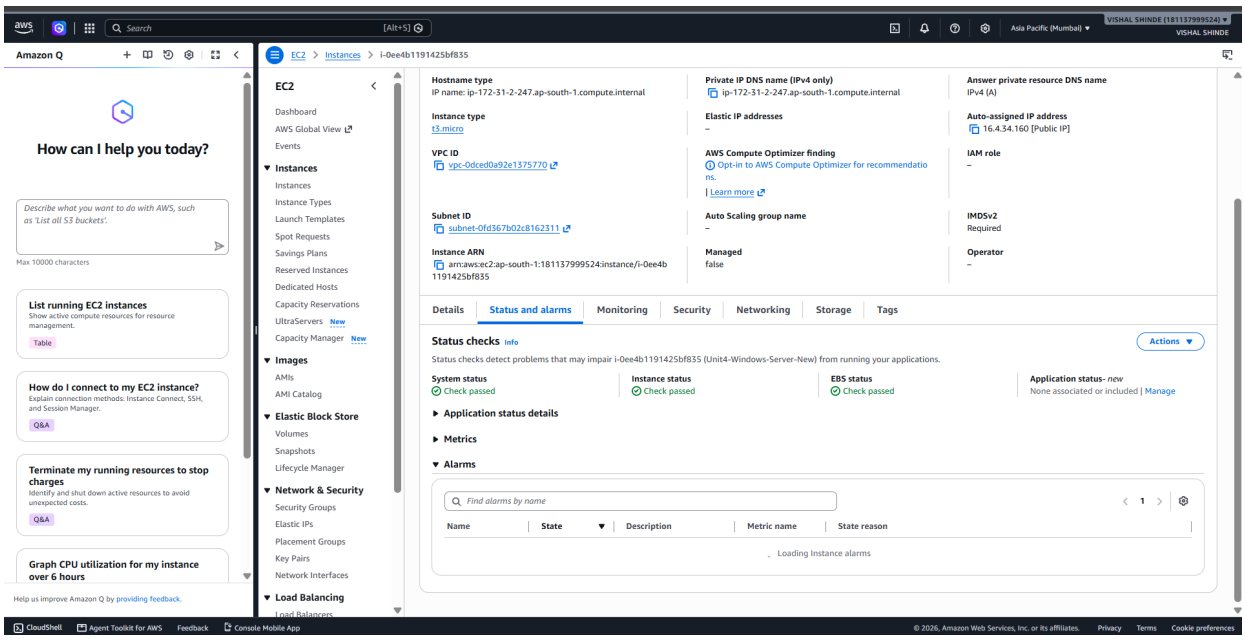
The following screenshots were captured during the practical and document the configuration and troubleshooting steps performed.

Evidence 1 — EC2 instance summary



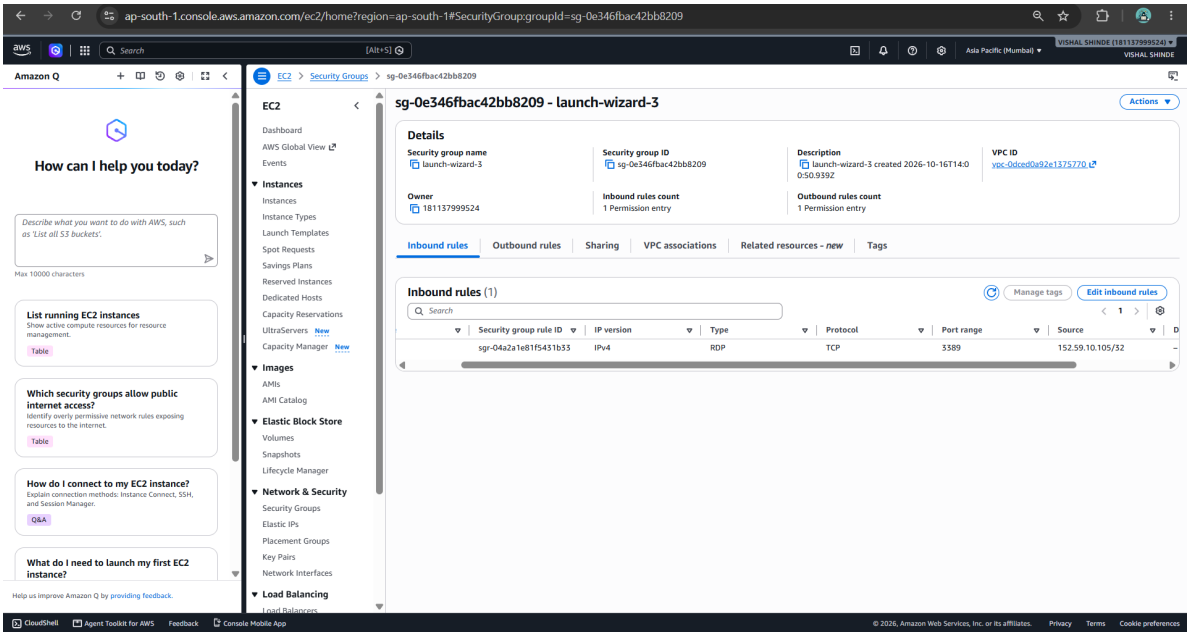
Shows the new Windows EC2 instance in Running state, public/private IP information, instance type, and network identifiers.

Evidence 2 — EC2 status checks



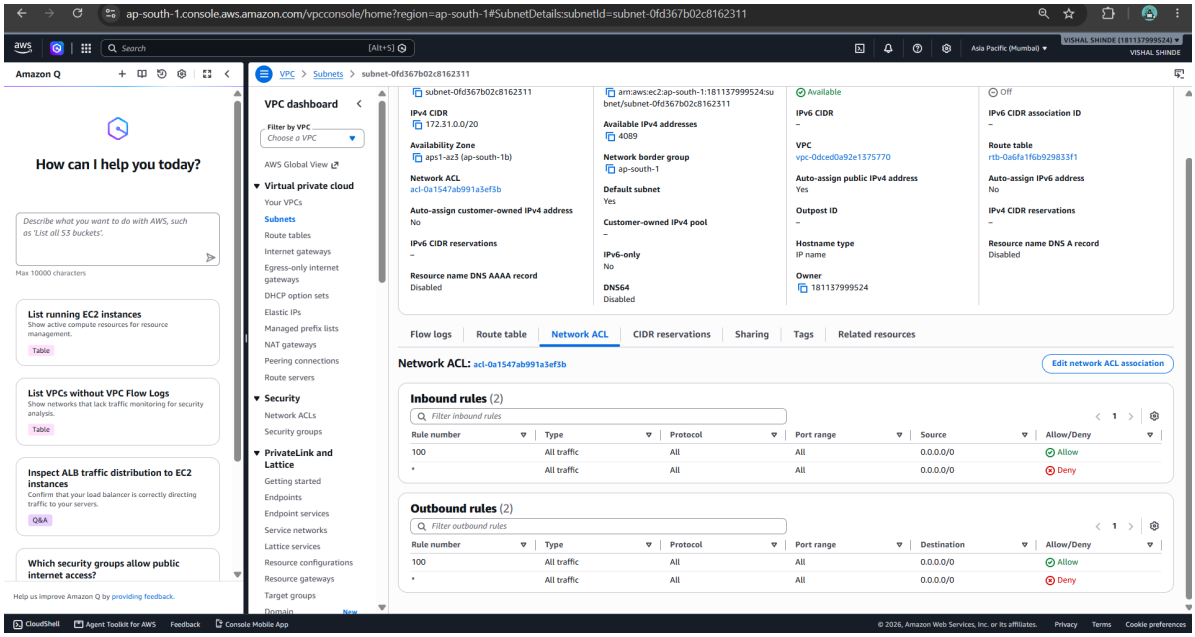
System, instance, and EBS status checks are shown as passed.

Evidence 3 — RDP security-group rule



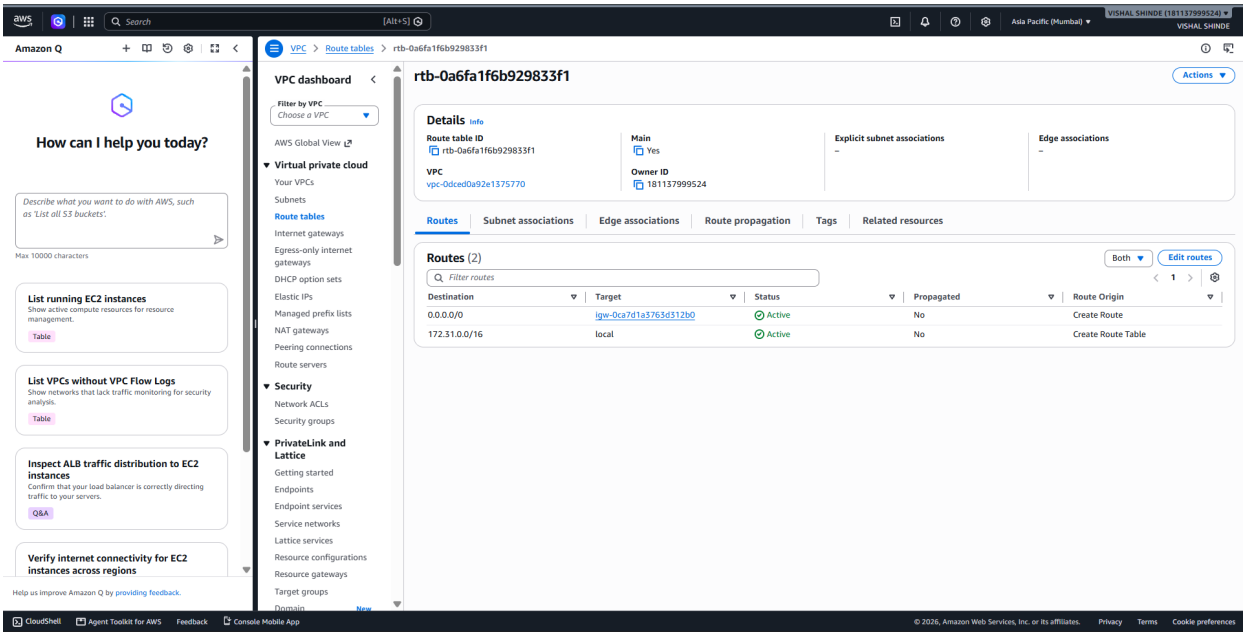
Inbound rule allows RDP over TCP port 3389 from a single public IP (/32), rather than from all IP addresses.

Evidence 4 — Subnet Network ACL



Inbound and outbound Network ACL rules allow traffic; the default rule denies traffic not matched by earlier rules.

Evidence 5 — Route table



The default route 0.0.0.0/0 points to an active Internet Gateway; the local VPC route is also active.

Evidence 6 — Initial RDP connectivity test

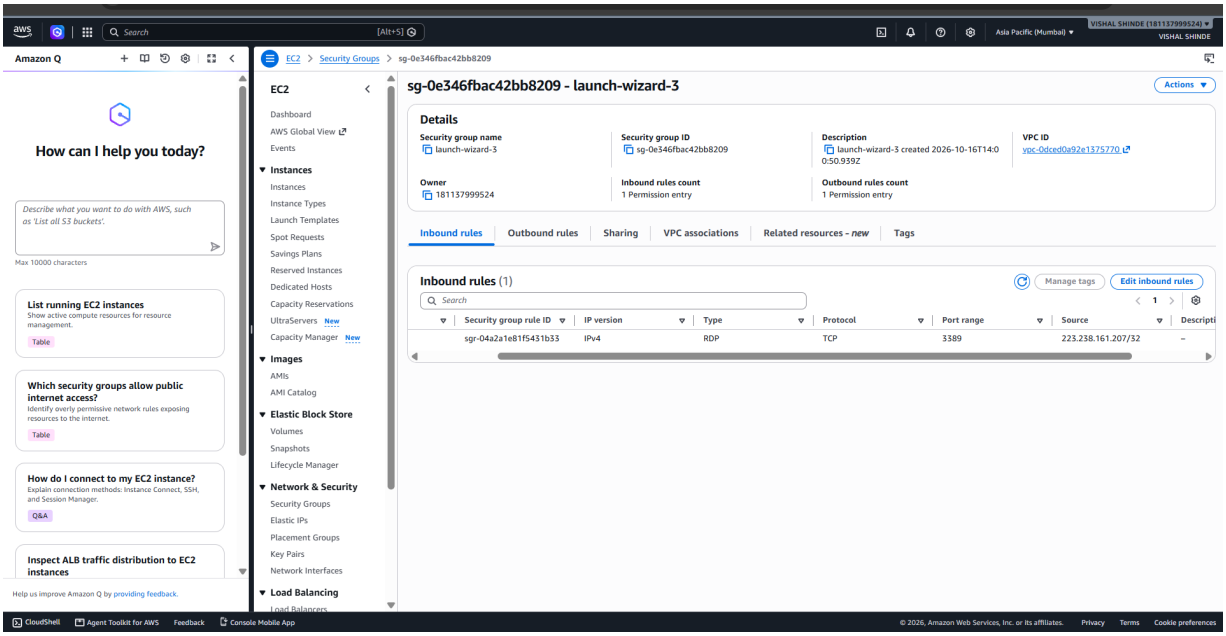
```
PS C:\WINDOWS\system32> Test-NetConnection -ComputerName 16.4.34.160 -Port 3389
WARNING: TCP connect to (16.4.34.160 : 3389) failed
WARNING: Ping to 16.4.34.160 failed with status: TimedOut


ComputerName            : 16.4.34.160
RemoteAddress           : 16.4.34.160
RemotePort              : 3389
InterfaceAlias          : WiFi 2
SourceAddress           : 10.56.223.6
PingSucceeded           : False
PingReplyDetails (RTT)  : 0 ms
TcpTestSucceeded        : False


PS C:\WINDOWS\system32>
```

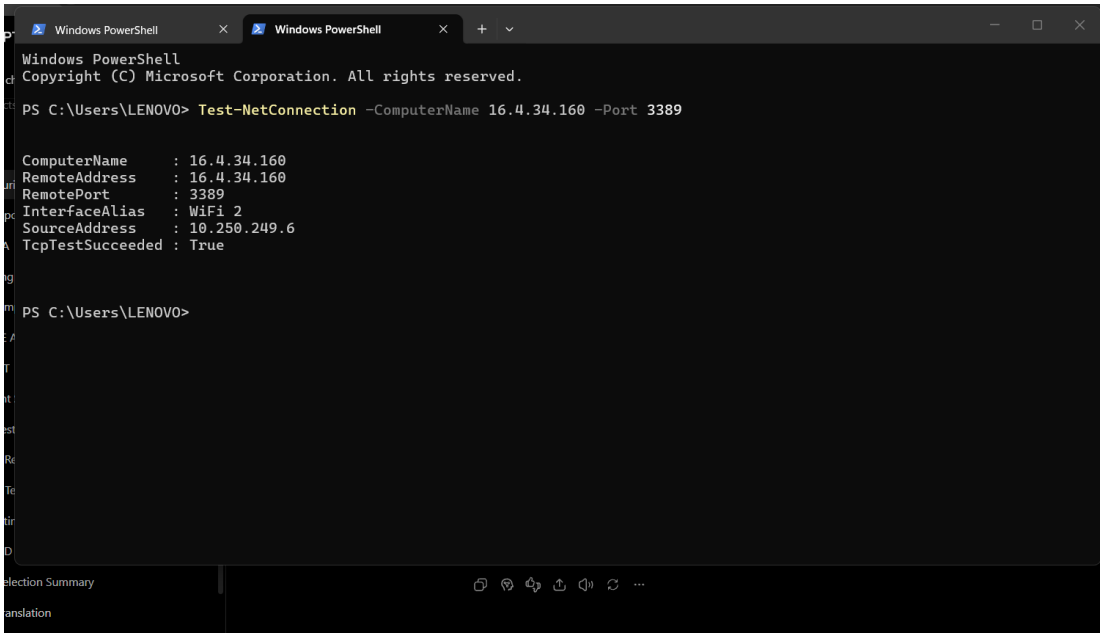
PowerShell Test-NetConnection to port 3389 returned TcpTestSucceeded: False on the initial network.

Evidence 7 — Updated RDP source IP



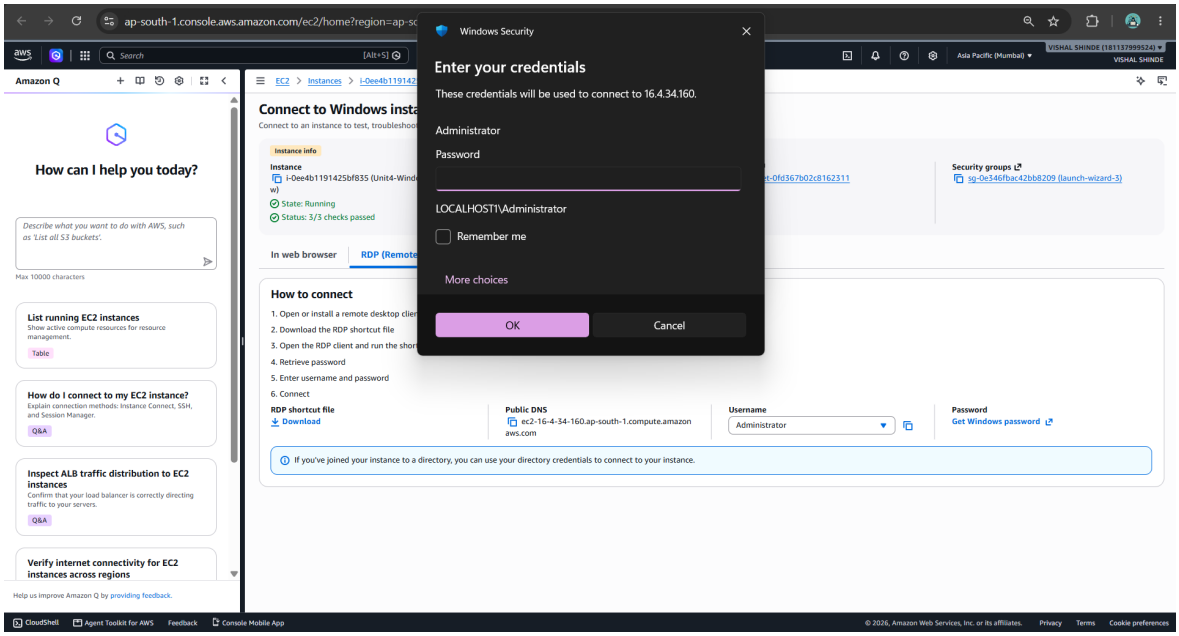
Security-group RDP source was changed to the hotspot public IP 223.238.161.207/32.

Evidence 8 — Successful TCP connectivity test



PowerShell Test-NetConnection returned TcpTestSucceeded: True while connected through the mobile hotspot.

Evidence 9 — Windows RDP credential prompt



Shows the Windows Security credential prompt for the EC2 public IP with Administrator selected. This is evidence of reaching the credential stage; it is not a screenshot of the final desktop.

Evidence limitation

A screenshot of the final Windows desktop was not included in the available images. The successful RDP session is recorded based on the user's confirmation; the included RDP screenshot shows the credential prompt, while the successful TCP connectivity screenshot independently confirms port 3389 was reachable.